

Master Thesis

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1 Introduction and scope of the project

In recent years, the role of propellers in the field of aircraft propulsion has become increasingly central, in particular the increase in the use of unmanned aerial vehicles, Unmanned Air Vehicles, and distributed electric propulsion, Distributed Electric Propulsion, has led to the development of small-diameter propellers. As stated in an article dealing with the future of the turboprop engine,¹ This technology is gaining increasing attention in the aviation market. Initially, the manufacturers' interest was to look for solutions to fly as high and fast as possible. The years between 1950 and 1970 were dubbed the "Jet Age", a veritable race for speed. Things changed when an oil crisis began in 1973 that led to an excessive increase in aviation costs, as higher speeds lead to high fuel consumption. Since then, sensitivity towards energy consumption and sustainability has increased. The fundamental parameter that is taken into consideration is the TSFC, Thrust Specific Fuel Consumption.

$$TSFC = \frac{\dot{m}_f}{T} \quad (1)$$

Where $\dot{m}_f$ is the mass flow rate of fuel consumed, while T is the thrust generated. The importance of this parameter is also closely linked to the maximum distance that can be covered; a lower cruising speed in fact allows for longer distances to be reached despite longer times.

Aircraft propellers therefore have various applications, they can be used for turboprop engines, for vehicles belonging to the VTOL, Vertical Take-Off and Landing category, or vehicles supported by DEP, Distributed Electric Propulsion.

The VTOL class in turn includes several uses, including urban mobility, the military sector, inspection or research drones. The main advantage of this technology is the absence of a runway along which acceleration is expected, a condition that becomes crucial especially in the military when timely interventions are planned, or landings in critical areas.

Distributed electric propulsion, on the other hand, consists of placing electrically driven, small-diameter propellers on the vehicle's wing to increase thrust. The aim here is to decrease, or even eliminate fuel consumption completely.

The scope of this thesis is to use one of the branches of artificial intelligence, namely Machine Learning, to reconstruct the propeller geometry from the performance data we want to obtain. Machine Learning² is defined as "the use and development of computer systems capable of learning and adapting

¹"Aircraft propellers-Is there a future?", P. Alves, M. Silvestre, 2020.

²ibm.com



Figure 1: VTOL, Flying Taxi Concept by Rolls-Royce

without the aid of specific instructions through the use of algorithms and statistical models to analyze patterns within a dataset.”

There are several types, but the one that will be used for this type of problem is the neural network. Neural networks are inspired by the analogy of biological connections, just as the human brain contains a large number of neurons that connect to each other via electrical impulses to perform all kinds of functions. Similarly, artificial neurons contain numerical information that is transmitted along a complex path.

Propeller reconstruction therefore consists of solving the inverse problem, from performance to geometry, in a process that comes very close to the concept of reverse engineering. In this work, the direct transition from geometry to performance will also be solved using different approaches. Initially, computational fluid dynamics will be used as a high-fidelity model to simulate the aerodynamic behavior of a given propeller, and later a low-fidelity model will be validated that allows for the use of fewer computational and time resources.

The geometries and their respective propeller performances were downloaded from the website of APC propeller ³, an American manufacturer of propellers for radio-controlled modeling, such as model airplanes, drones, or helicopters. They offer different solutions depending on the type of drive used: Combustion Engine, Electric Motors or drone propellers. Adequate neural network training requires the provision of a sufficient amount of data. The APC site allows for the download of geometries and performance of nearly 500 different helices with diameters ranging from approximately 15cm to 60cm.

The CFD was carried out using the commercial software STAR-CCM+⁴, XROTOR⁵ was chosen as the low-fidelity model, implemented with XFOIL, two open source programs.

³apcprop.com

⁴starccm.smartcae.com

⁵web.mit.edu